Grade 10 Term 1 Lesson 1 Practice Activity 1
Decision Alternatives, Events, Consequences, States, and Payoff Tables

Contents

Evaluated criteria

C1 Characterises decision-making problems under uncertainty by identifying decision alternatives, states of nature, consequences, and the economic or financial context of the decision.

C2 Organises decision problems using payoff tables, placing alternatives, states of nature, and payoffs in a clear structure for comparison.

1 Introduction

Problems:

A decision problem under uncertainty describes a choice that must be made before an uncertain event is known. The decision-maker chooses one *alternative*; later, the uncertain event produces one *state of nature*. A *consequence* describes what happens for that alternative–state combination, and a *payoff* is the numerical value assigned to the consequence. Thus, consequences describe outcomes in words, while payoffs express their economic or financial values in numbers.

Let

$$A_1, A_2, \ldots, A_m$$

be the available decision alternatives, where i identifies a particular alternative, and let

$$S_1, S_2, \ldots, S_n$$

be the possible states of nature, where j identifies a particular state. The decision-maker chooses one A_i, and the uncertain event subsequently produces one S_j. The pair (A_i, S_j) produces one consequence, whose numerical payoff is $P_{ay}(A_i, S_j)$. Alternatives form the rows of a payoff table and states form its columns, so

$$\mathbf{P} = \begin{pmatrix} P_{ay}(A_1, S_1) & P_{ay}(A_1, S_2) & \cdots & P_{ay}(A_1, S_n) \\ P_{ay}(A_2, S_1) & P_{ay}(A_2, S_2) & \cdots & P_{ay}(A_2, S_n) \\ \vdots & \vdots & \ddots & \vdots \\ P_{ay}(A_m, S_1) & P_{ay}(A_m, S_2) & \cdots & P_{ay}(A_m, S_n) \end{pmatrix}.$$

In particular, $P_{ay}(A_i, S_j)$ means the payoff obtained when alternative A_i is selected and state S_j occurs.

More advanced problems use the following payoff components.

Symbol	Meaning
$q(A_i, S_j)$	Number of items for A_i under S_j
$v(A_i)$	Variable payoff per item determined by A_i
$v(S_j)$	Variable payoff per item determined by S_j
$f(A_i)$	Fixed payoff determined by A_i
$f(S_j)$	Fixed payoff determined by S_j
$P_{ay}(A_i, S_j)$	Total payoff for the pair (A_i, S_j)

The complete four-component structure is

$$P_{ay}(A_i, S_j) = q(A_i, S_j)\big[v(A_i) + v(S_j)\big] + f(A_i) + f(S_j).$$

Here, $q(A_i, S_j)v(A_i)$ is the variable payoff generated by the alternative-based rate; $q(A_i, S_j)v(S_j)$ is the variable payoff generated by the state-based rate; $f(A_i)$ is the alternative-based fixed component; and $f(S_j)$ is the state-based fixed component. Adding all applicable components gives $P_{ay}(A_i, S_j)$. A component whose argument is A_i changes by alternative (and therefore by row), one whose argument is S_j changes by state (and therefore by column), and one with both arguments, such as $q(A_i, S_j)$, can change for every cell.

Simpler structures occur when components are absent; an absent component can be treated as zero. For example, $P_{ay}(A_i, S_j) = q(A_i, S_j)v(A_i)$ uses only an alternative-based variable component, while $P_{ay}(A_i, S_j) = f(A_i) + f(S_j)$ uses only the two fixed components. Directly supplied payoffs need no decomposition: when $P_{ay}(A_i, S_j)$ is already given, it is placed directly in its payoff-table cell. The direct-payoff form and the complete four-component form therefore describe the same table structure; they differ only in whether the cell value is supplied or calculated.

A reusable procedure is:

1. Identify the decision and its economic or financial context.
2. List the available alternatives A_i.
3. Identify the uncertain event.

4. List its possible states S_j.
5. Determine the consequence for every pair (A_i, S_j).
6. Identify which payoff values depend on the alternative and which depend on the state.
7. Calculate each $P_{ay}(A_i, S_j)$ using the appropriate formula, unless it is supplied directly.
8. Place alternatives in rows and states in columns.
9. Check that every alternative–state pair has exactly one payoff.
10. Check that all quantities and payoff components use compatible units.

The purpose is to convert a verbal decision situation into a complete, comparable payoff table. At this stage, the purpose is not to select the best alternative.

1.1 Introductory General Problem (C1–C2)

A decision-maker chooses between generic alternatives A_1 and A_2. After the choice, one uncertain event produces either state S_1 or state S_2. The payoffs, in thousand USD, are supplied directly:

$$P_{ay}(A_1, S_1) = 10, \quad P_{ay}(A_1, S_2) = 4, \quad P_{ay}(A_2, S_1) = 7, \quad P_{ay}(A_2, S_2) = 8.$$

The arguments A_i and S_j identify the alternative (the row) and state (the column), respectively. The four combinations are

$$(A_1, S_1), \quad (A_1, S_2), \quad (A_2, S_1), \quad (A_2, S_2).$$

Questions/Tasks.

C1 Interpreting decision alternatives, events, consequences, and states.

C2 Building the payoff table.

C1 - Interpreting decision alternatives, events, consequences, and states.

Element	Description	
Alternatives	**A1:** A_1	**A2:** A_2
Statesofnature	**S1:** S_1	**S2:** S_2
Events	The uncertain event that produces either S_1 or S_2	
Consequences	The consequence and its payoff in thousand USD for each pair (A_i, S_j)	

C2 - Building the payoff table.

Alternative	S_1	S_2
A_1	10	4
A_2	7	8

Because $P_{ay}(A_1, S_1)$ belongs to row A_1 and column S_1, 10 goes in the first payoff cell. Similarly, $P_{ay}(A_1, S_2) = 4$ goes in row A_1, column S_2; $P_{ay}(A_2, S_1) = 7$ goes in row A_2, column S_1; and $P_{ay}(A_2, S_2) = 8$ goes in row A_2, column S_2.

1.2 Advanced General Problem (C1–C2)

A decision-maker chooses A_1 or A_2 before an uncertain event produces S_1 or S_2. All monetary values below are in thousand USD, with variable values stated per item:

$$v(A_1) = 2, \quad v(A_2) = 3, \qquad f(A_1) = 5, \quad f(A_2) = 6,$$

$$v(S_1) = 1, \quad v(S_2) = 2, \qquad f(S_1) = 1, \quad f(S_2) = 2.$$

The alternative–state quantities are:

Alternative	S_1	S_2
A_1	$q(A_1, S_1) = 2$	$q(A_1, S_2) = 3$
A_2	$q(A_2, S_1) = 4$	$q(A_2, S_2) = 5$

Questions/Tasks.

C1 Interpreting decision alternatives, events, consequences, and states.

C2 Building the payoff table.

C1 - Interpreting decision alternatives, events, consequences, and states.

Element	Description	
Alternatives	**A1:** A_1	**A2:** A_2
Statesofnature	**S1:** S_1	**S2:** S_2
Events	The uncertain event that produces either S_1 or S_2	
Consequences	The total payoff in thousand USD for each pair (A_i, S_j)	

C2 - Building the payoff table.

Every cell uses the complete formula

$$P_{ay}(A_i, S_j) = q(A_i, S_j)\,(v(A_i) + v(S_j)) + f(A_i) + f(S_j).$$

The components whose argument is A_i change by alternative and therefore by row. The components whose argument is S_j change by state and therefore by column. The quantity $q(A_i, S_j)$ depends on both arguments and can change in every cell.

Payoff	Complete numerical substitution	Result
$P_{ay}(A_1, S_1)$	$q(A_1, S_1)(v(A_1) + v(S_1)) + f(A_1) + f(S_1) = 2(2 + 1) + 5 + 1$	12
$P_{ay}(A_1, S_2)$	$q(A_1, S_2)(v(A_1) + v(S_2)) + f(A_1) + f(S_2) = 3(2 + 2) + 5 + 2$	19
$P_{ay}(A_2, S_1)$	$q(A_2, S_1)(v(A_2) + v(S_1)) + f(A_2) + f(S_1) = 4(3 + 1) + 6 + 1$	23
$P_{ay}(A_2, S_2)$	$q(A_2, S_2)(v(A_2) + v(S_2)) + f(A_2) + f(S_2) = 5(3 + 2) + 6 + 2$	33

Each line displays separately the alternative–state quantity, alternative-based variable value, state-based variable value, alternative-based fixed value, and state-based fixed value before they are combined.

Alternative	S_1	S_2
A_1	12	19
A_2	23	33

2 Direct Payoff Models

Problems:

2.1 Direct

Problems:

2.1. Local Café Investment (C1–C2)
2.1. Green Energy Production Mix (C1–C2)
2.1. Global Shipping Network Design (C1–C2)

Local Café Investment (C1–C2)

A small café is deciding whether to launch a new line of artisan desserts. The transaction is daily café sales, where profit is earned from each dessert plate sold. The owner has two options:

1. **Launch the dessert line**, with profit outcomes depending on foot traffic:
   i) High traffic → 42 thousand USD
   ii) Low traffic → 8 thousand USD

2. **Keep the current menu**, with profit outcomes based on historical data:
   i) High traffic → 28 thousand USD
   ii) Low traffic → 20 thousand USD

Questions/Tasks.

C1 Interpreting decision alternatives, events, consequences, and states.

C2 Building the payoff table.

C1 - Interpreting decision alternatives, events, consequences, and states.

Element	Description	
Alternatives	**A1:** Launch dessert line	**A2:** Keep current menu
Statesofnature	**S1:** High traffic	**S2:** Low traffic
Events	Actual foot traffic after the decision	
Consequences	Profit in thousand USD for each alternative–state pair	

C2 - Building the payoff table.

Alternative	High traffic	Low traffic
Launch dessert line	42	8
Keep menu	28	20

Green Energy Production Mix (C1–C2)

A renewable-energy company must select an electricity generation mix. The transaction is annual electricity sales, where profit is earned per megawatt-hour generated and sold.

Three weather states are possible:

1. Windy
2. Sunny
3. Cloudy

The company is comparing three strategies:

1. Build wind turbines
2. Build solar farms
3. Build a balanced hybrid system

Modeled profits in thousand USD:

1. Wind turbines → 90 (Windy), 45 (Sunny), 30 (Cloudy)
2. Solar farms → 35 (Windy), 95 (Sunny), 25 (Cloudy)
3. Hybrid system → 70 (Windy), 65 (Sunny), 60 (Cloudy)

Questions/Tasks.

C1 Interpreting decision alternatives, events, consequences, and states.

C2 Building the payoff table.

C1 - Interpreting decision alternatives, events, consequences, and states.

Element	Description		
Alternatives	**A1:** Wind turbines	**A2:** Solar farms	**A3:** Hybrid system
Statesofnature	**S1:** Windy	**S2:** Sunny	**S3:** Cloudy
Events	Weather outcome affecting generation		
Consequences	Profit in thousand USD for each strategy–weather combination		

C2 - Building the payoff table.

Strategy	Windy	Sunny	Cloudy
Wind turbines	90	45	30
Solar farms	35	95	25
Hybrid system	70	65	60

Global Shipping Network Design (C1–C2)

A multinational logistics firm is choosing between two shipping network designs for long-term freight contracts. The transaction is global freight services, where profit is earned per contract delivered on schedule.

Management is considering:

1. **Centralized mega-hub network**
2. **Regional multi-hub network**

The firm models four trade conditions:

1. Trade boom
2. Stable trade
3. Moderate disruptions
4. Severe disruptions

Modeled profits in thousand USD are:

1. Centralized network $\rightarrow 220, 140, 40, -30$
2. Regional network $\rightarrow 180, 160, 110, 60$

Questions/Tasks.

C1 Interpreting decision alternatives, events, consequences, and states.

C2 Building the payoff table.

C1 - Interpreting decision alternatives, events, consequences, and states.

Element	Description			
Alternatives	**A1:** Centralized mega-hub	**A2:** Regional multi-hub		
Statesofnature	**S1:** Boom	**S2:** Stable	**S3:** Moderate disruptions	**S4:** Severe disruptions
Events	Trade conditions realized during the planning horizon			
Consequences	Profit in thousand USD for each network–state pair			

C2 - Building the payoff table.

Network	Boom	Stable	Moderate disruptions	Severe disruptions
Centralized mega-hub	220	140	40	-30
Regional multi-hub	180	160	110	60

3 One-Component Models

Problems:

3.1 Alternative-Based Variable

Problems:

Community Bakery Lunch Boxes (C1–C2)

A community bakery is bidding on office lunch-box contracts. The transaction is meal box sales, where profit is earned per lunch box delivered.

The bakery can choose one pricing strategy:

1. **Standard packaging** with a profit of 0.04 thousand USD per lunch box
2. **Premium packaging** with a profit of 0.07 thousand USD per lunch box

Modeled number of lunch boxes:

1. Standard packaging → 900 (High), 550 (Low)
2. Premium packaging → 650 (High), 250 (Low)

Demand states:

1. High office demand
2. Low office demand

Questions/Tasks.

C1 Interpreting decision alternatives, events, consequences, and states.

C2 Building the payoff table.

C1 - Interpreting decision alternatives, events, consequences, and states.

Element	Description	
Alternatives	**A1:** Standard packaging	**A2:** Premium packaging
Statesofnature	**S1:** High demand	**S2:** Low demand
Events	Actual number of lunch-box contracts realized	
Consequences	Profit in thousand USD, based on boxes × profit per box	

C2 - Building the payoff table. Calculation table in thousand USD:

$$\text{Payoff} = \text{number of lunch boxes} \times \text{profit per lunch box}$$

Alternative	High demand	Low demand
Standard packaging	$900 \times 0.04 = 36$	$550 \times 0.04 = 22$
Premium packaging	$650 \times 0.07 = 45.5$	$250 \times 0.07 = 17.5$

Alternative	High demand	Low demand
Standard packaging	36	22
Premium packaging	45.5	17.5

Rural Internet Subscriptions (C1–C2)

A rural internet provider is selecting a subscription plan mix. The transaction is monthly broadband subscriptions, where profit is earned per subscription sold.

Decision alternatives:

1. **Basic plan** with profit of 0.03 thousand USD per subscription
2. **Plus plan** with profit of 0.05 thousand USD per subscription
3. **Pro plan** with profit of 0.08 thousand USD per subscription

Modeled subscriptions sold:

1. Basic → 1400 (High), 900 (Low)
2. Plus → 1000 (High), 700 (Low)
3. Pro → 600 (High), 300 (Low)

Enrollment states:

1. High enrollment
2. Low enrollment

Questions/Tasks.

C1 Interpreting decision alternatives, events, consequences, and states.

C2 Building the payoff table.

C1 - Interpreting decision alternatives, events, consequences, and states.

Element	Description		
Alternatives	**A1:** Basic plan	**A2:** Plus plan	**A3:** Pro plan
Statesofnature	**S1:** High enrollment	**S2:** Low enrollment	
Events	Actual subscription uptake after plan selection		
Consequences	Profit in thousand USD based on subscriptions × profit per subscription		

C2 - Building the payoff table. Calculation table in thousand USD:

$$\text{Payoff} = \text{subscriptions sold} \times \text{profit per subscription}$$

Alternative	High enrollment	Low enrollment
Basic plan	$1400 \times 0.03 = 42$	$900 \times 0.03 = 27$
Plus plan	$1000 \times 0.05 = 50$	$700 \times 0.05 = 35$
Pro plan	$600 \times 0.08 = 48$	$300 \times 0.08 = 24$

Alternative	High enrollment	Low enrollment
Basic plan	42	27
Plus plan	50	35
Pro plan	48	24

3.2 State-Based Variable

Problems:
3.2. Coffee Export Routes (C1–C2)
3.2. Lake Ferry Tourism (C1–C2)

Coffee Export Routes (C1–C2)

A coffee exporter must choose a shipping route. The transaction is exporting coffee shipments, where profit is earned per shipment delivered.

Decision alternatives:

1. **Route through Port A**
2. **Route through Port B**

Exchange-rate states:

1. Favorable rate
2. Unfavorable rate

Profit per shipment depends on the exchange rate:

1. Favorable $\rightarrow$ 6 thousand USD per shipment
2. Unfavorable $\rightarrow$ 2 thousand USD per shipment

Modeled shipments:

1. Port A $\rightarrow$ 40 (Favorable), 28 (Unfavorable)
2. Port B $\rightarrow$ 32 (Favorable), 30 (Unfavorable)

Questions/Tasks.

C1 Interpreting decision alternatives, events, consequences, and states.

C2 Building the payoff table.

C1 - Interpreting decision alternatives, events, consequences, and states.

Element	Description	
Alternatives	**A1:** Port A route	**A2:** Port B route
Statesofnature	**S1:** Favorable rate	**S2:** Unfavorable rate
Events	Exchange-rate level that affects profit per shipment	
Consequences	Profit in thousand USD based on shipments × profit per shipment	

C2 - Building the payoff table. Calculation table in thousand USD:

$$\text{Payoff} = \text{shipments} \times \text{profit per shipment}$$

Route	Favorable rate	Unfavorable rate
Port A	$40 \times 6 = 240$	$28 \times 2 = 56$
Port B	$32 \times 6 = 192$	$30 \times 2 = 60$

Route	Favorable rate	Unfavorable rate
Port A	240	56
Port B	192	60

Lake Ferry Tourism (C1–C2)

A lake tourism operator is choosing boat sizes for sightseeing tours. The transaction is tour ticket sales, where profit is earned per ticket sold.

Alternatives:

1. **Operate small ferries**
2. **Operate large ferries**

Seasonal states:

1. Peak season
2. Normal season
3. Rainy season

Profit per ticket depends on the season:

1. Peak → 0.05 thousand USD
2. Normal → 0.04 thousand USD
3. Rainy → 0.03 thousand USD

Modeled tickets sold:

1. Small ferries → 2200 (Peak), 1600 (Normal), 1000 (Rainy)
2. Large ferries → 2600 (Peak), 1800 (Normal), 800 (Rainy)

Questions/Tasks.

C1 Interpreting decision alternatives, events, consequences, and states.

C2 Building the payoff table.

C1 - Interpreting decision alternatives, events, consequences, and states.

Element	Description		
Alternatives	**A1:** Small ferries	**A2:** Large ferries	
Statesofnature	**S1:** Peak	**S2:** Normal	**S3:** Rainy
Events	Tourism season and ticket price level		
Consequences	Profit in thousand USD based on tickets × profit per ticket		

C2 - Building the payoff table. Calculation table in thousand USD:

$$\text{Payoff} = \text{tickets sold} \times \text{profit per ticket}$$

Alternative	Peak	Normal	Rainy
Small ferries	$2200 \times 0.05 = 110$	$1600 \times 0.04 = 64$	$1000 \times 0.03 = 30$
Large ferries	$2600 \times 0.05 = 130$	$1800 \times 0.04 = 72$	$800 \times 0.03 = 24$

Alternative	Peak	Normal	Rainy
Small ferries	110	64	30
Large ferries	130	72	24

3.3 Alternative-Based Fixed

Problems:

School Meal Service Contracts (C1–C2)

A school cafeteria provider is choosing a meal-service contract for a short academic program. The transaction is a fixed service contract, where the recorded payoff is tied only to the selected contract type.

Decision alternatives with fixed payoff components:

1. **Basic meal contract** → 18 thousand USD
2. **Extended meal contract** → 32 thousand USD

Participation states:

1. Regular participation
2. High participation

Questions/Tasks.

C1 Interpreting decision alternatives, events, consequences, and states.

C2 Building the payoff table.

C1 - Interpreting decision alternatives, events, consequences, and states.

Element	Description	
Alternatives	**A1:** Basic meal	**A2:** Extended meal
Statesofnature	**S1:** Regular participation	**S2:** High participation
Events	Participation level observed after the contract choice	
Consequences	Fixed payoff in thousand USD tied only to the selected contract	

C2 - Building the payoff table.

Alternative	Regular participation	High participation
Basic meal	18	18
Extended meal	32	32

Mobile Clinic Service Contracts (C1–C2)

A health nonprofit is choosing a service contract for mobile clinics. The transaction is a fixed service contract, where the recorded payoff is tied only to the selected contract type.

Decision alternatives with fixed payoff components:

1. **Basic outreach contract** → 24 thousand USD
2. **Extended outreach contract** → 38 thousand USD
3. **Full regional contract** → 52 thousand USD

Community-need states:

1. Regular need
2. High need

Questions/Tasks.

C1 Interpreting decision alternatives, events, consequences, and states.

C2 Building the payoff table.

C1 - Interpreting decision alternatives, events, consequences, and states.

Element	Description		
Alternatives	**A1:** Basic outreach	**A2:** Extended outreach	**A3:** Full regional
Statesofnature	**S1:** Regular need	**S2:** High need	
Events	Community-need level observed after the contract choice		
Consequences	Fixed payoff in thousand USD tied only to the selected contract		

C2 - Building the payoff table. Calculation table in thousand USD:

$$\text{Payoff} = \text{contract fixed payoff}$$

Alternative	Regular need	High need
Basic outreach	24	24
Extended outreach	38	38
Full regional	52	52

Alternative	Regular need	High need
Basic outreach	24	24
Extended outreach	38	38
Full regional	52	52

3.4 State-Based Fixed

Problems:
3.4. Neighborhood Shelter Support (C1–C2)
3.4. Community Hall Emergency Rentals (C1–C2)

Neighborhood Shelter Support (C1–C2)

A municipal office is deciding how to prepare a temporary shelter service for short emergency responses. The transaction is public-service shelter support, where the recorded payoff is a fixed municipal support amount tied only to the emergency state.

Decision alternatives:

1. **Open the school gym**
2. **Open the sports center**

Emergency states with fixed payoff components:

1. Minor emergency → 12 thousand USD
2. Major emergency → 30 thousand USD

Questions/Tasks.

C1 Interpreting decision alternatives, events, consequences, and states.

C2 Building the payoff table.

C1 - Interpreting decision alternatives, events, consequences, and states.

Element	Description	
Alternatives	**A1:** School gym	**A2:** Sports center
Statesofnature	**S1:** Minor emergency	**S2:** Major emergency
Events	Emergency level realized after the shelter decision	
Consequences	Fixed payoff in thousand USD tied only to the emergency state	

C2 - Building the payoff table.

Alternative	Minor emergency	Major emergency
School gym	12	30
Sports center	12	30

Community Hall Emergency Rentals (C1–C2)

A city council is choosing how to reserve community halls for emergency use. The transaction is public-service facility rental, where the recorded payoff is a fixed municipal support amount tied only to the emergency state.

Decision alternatives:

1. **Reserve one central hall**
2. **Reserve two neighborhood halls**

Emergency states with fixed payoff components:

1. Mild emergency → 18 thousand USD
2. Serious emergency → 35 thousand USD
3. Major emergency → 50 thousand USD

Questions/Tasks.

C1 Interpreting decision alternatives, events, consequences, and states.

C2 Building the payoff table.

C1 - Interpreting decision alternatives, events, consequences, and states.

Element	Description		
Alternatives	**A1:** One central hall	**A2:** Two neighborhood halls	
Statesofnature	**S1:** Mild emergency	**S2:** Serious emergency	**S3:** Major emergency
Events	Emergency level realized after the reservation decision		
Consequences	Fixed payoff in thousand USD tied only to the emergency state		

C2 - Building the payoff table. Calculation table in thousand USD:

$$\text{Payoff} = \text{emergency-state fixed payoff}$$

Alternative	Mild emergency	Serious emergency	Major emergency
One central hall	18	35	50
Two neighborhood halls	18	35	50

Alternative	Mild emergency	Serious emergency	Major emergency
One central hall	18	35	50
Two neighborhood halls	18	35	50

4 Two-Component Models

Problems:

4.1 Alternative-Based Variable Plus State-Based Variable

Problems:

Recycling Collection Contract (C1–C2)

A municipal contractor chooses a collection fleet for a two-month recycling service contract.

Decision alternatives

1. Electric fleet
2. Diesel fleet

States of nature

1. High recovery
2. Low recovery

Alternative-based variable payoff per item $v(A_i)$

1. Electric fleet: 0.6 thousand USD
2. Diesel fleet: 0.4 thousand USD

State-based variable payoff per item $v(S_j)$

1. High recovery: 0.3 thousand USD
2. Low recovery: 0.1 thousand USD

Modeled number of items $q(A_i, S_j)$

1. Electric fleet: 80 under High recovery, 55 under Low recovery
2. Diesel fleet: 95 under High recovery, 65 under Low recovery

Questions/Tasks.

C1 Interpreting decision alternatives, events, consequences, and states.

C2 Building the payoff table.

C1 - Interpreting decision alternatives, events, consequences, and states.

Element	Description	
Alternatives	**A1:** Electric fleet	**A2:** Diesel fleet
Statesofnature	**S1:** High recovery	**S2:** Low recovery
Events	The realized state affecting the transaction	
Consequences	Payoff in thousand USD for each alternative–state pair	

C2 - Building the payoff table. The applicable payoff formula is

$$P_{ay}(A_i, S_j) = q(A_i, S_j)\,(v(A_i) + v(S_j))\,.$$

Detailed calculations (thousand USD):

Alternative	High recovery	Low recovery
Electric fleet	$80 \times (0.6 + 0.3) = 72$	$55 \times (0.6 + 0.1) = 38.5$
Diesel fleet	$95 \times (0.4 + 0.3) = 66.5$	$65 \times (0.4 + 0.1) = 32.5$

The simplified payoff table is

Alternative	High recovery	Low recovery
Electric fleet	72	38.5
Diesel fleet	66.5	32.5

Museum Ticket Platform (C1–C2)

A museum consortium selects a ticket platform; the transaction is batches of one thousand admissions processed.

Decision alternatives

1. Mobile platform
2. Kiosk platform
3. Hybrid platform

States of nature

1. Peak season

2. Regular season

Alternative-based variable payoff per item $v(A_i)$

1. Mobile platform: 0.5 thousand USD
2. Kiosk platform: 0.4 thousand USD
3. Hybrid platform: 0.4 thousand USD

State-based variable payoff per item $v(S_j)$

1. Peak season: 0.2 thousand USD
2. Regular season: 0.1 thousand USD

Modeled number of items $q(A_i, S_j)$

1. Mobile platform: 70 under Peak season, 45 under Regular season
2. Kiosk platform: 60 under Peak season, 50 under Regular season
3. Hybrid platform: 65 under Peak season, 55 under Regular season

Questions/Tasks.

C1 Interpreting decision alternatives, events, consequences, and states.

C2 Building the payoff table.

C1 - Interpreting decision alternatives, events, consequences, and states.

Element	Description		
Alternatives	**A1:** Mobile platform	**A2:** Kiosk platform	**A3:** Hybrid platform
Statesofnature	**S1:** Peak season	**S2:** Regular season	
Events	The realized state affecting the transaction		
Consequences	Payoff in thousand USD for each alternative–state pair		

C2 - Building the payoff table. The applicable payoff formula is

$$P_{ay}(A_i, S_j) = q(A_i, S_j)\,(v(A_i) + v(S_j))\,.$$

Detailed calculations (thousand USD):

Alternative	Peak season	Regular season
Mobile platform	$70 \times (0.5 + 0.2) = 49$	$45 \times (0.5 + 0.1) = 27$
Kiosk platform	$60 \times (0.4 + 0.2) = 36$	$50 \times (0.4 + 0.1) = 25$
Hybrid platform	$65 \times (0.4 + 0.2) = 39$	$55 \times (0.4 + 0.1) = 27.5$

The simplified payoff table is

Alternative	Peak season	Regular season
Mobile platform	49	27
Kiosk platform	36	25
Hybrid platform	39	27.5

4.2 Alternative-Based Variable Plus Alternative-Based Fixed

Problems:

4.2. Local Printing Contracts (C1–C2)

4.2. Cold-Storage Contracts (C1–C2)

Local Printing Contracts (C1–C2)

A local printing shop is choosing a contract format for school material packets. The transaction is printing packets, where profit is earned per packet printed plus a fixed profit associated with the chosen printing setup.

Decision alternatives:

1. **Small printer setup** (fixed profit 12 thousand USD, profit 0.05 thousand USD per packet)
2. **Digital printer setup** (fixed profit 20 thousand USD, profit 0.04 thousand USD per packet)

Modeled packets printed:

1. Small printer → 500 (High), 300 (Low)
2. Digital printer → 700 (High), 400 (Low)

Demand states:

1. High packet demand
2. Low packet demand

Questions/Tasks.

C1 Interpreting decision alternatives, events, consequences, and states.

C2 Building the payoff table.

C1 - Interpreting decision alternatives, events, consequences, and states.

Element	Description	
Alternatives	**A1:** Small printer	**A2:** Digital printer
Statesofnature	**S1:** High demand	**S2:** Low demand
Events	Packet demand realized after the printing setup decision	
Consequences	Profit in thousand USD = packets × profit per packet + fixed profit	

C2 - Building the payoff table. Calculation table in thousand USD:

$$\text{Payoff} = \text{packets printed} \times \text{profit per packet} + \text{setup fixed profit}$$

Alternative	High demand	Low demand
Small printer	$500 \times 0.05 + 12 = 37$	$300 \times 0.05 + 12 = 27$
Digital printer	$700 \times 0.04 + 20 = 48$	$400 \times 0.04 + 20 = 36$

Alternative	High demand	Low demand
Small printer	37	27
Digital printer	48	36

Cold-Storage Contracts (C1–C2)

A logistics company is leasing cold-storage space for agricultural pallets. The transaction is storage contracts, where profit is earned per pallet stored plus a fixed profit associated with the chosen facility.

Decision alternatives:

1. **Small facility** (fixed profit 40 thousand USD, profit 0.18 thousand USD per pallet)
2. **Medium facility** (fixed profit 25 thousand USD, profit 0.12 thousand USD per pallet)
3. **Large facility** (fixed profit 10 thousand USD, profit 0.08 thousand USD per pallet)

Modeled pallets stored:

1. Small → 300 (High), 220 (Low)
2. Medium → 600 (High), 420 (Low)
3. Large → 900 (High), 700 (Low)

Demand states:

1. High storage demand
2. Low storage demand

Questions/Tasks.

C1 Interpreting decision alternatives, events, consequences, and states.

C2 Building the payoff table.

C1 - Interpreting decision alternatives, events, consequences, and states.

Element	Description		
Alternatives	**A1:** Small facility	**A2:** Medium facility	**A3:** Large facility
Statesofnature	**S1:** High demand	**S2:** Low demand	
Events	Storage demand realized after the lease decision		
Consequences	Profit in thousand USD = pallets × profit per pallet + fixed profit		

C2 - Building the payoff table. Calculation table in thousand USD:

$$\text{Payoff} = \text{pallets stored} \times \text{profit per pallet} + \text{facility fixed profit}$$

Alternative	High demand	Low demand
Small facility	$300 \times 0.18 + 40 = 94$	$220 \times 0.18 + 40 = 79.6$
Medium facility	$600 \times 0.12 + 25 = 97$	$420 \times 0.12 + 25 = 75.4$
Large facility	$900 \times 0.08 + 10 = 82$	$700 \times 0.08 + 10 = 66$

Alternative	High demand	Low demand
Small facility	94	79.6
Medium facility	97	75.4
Large facility	82	66

4.3 Alternative-Based Variable Plus State-Based Fixed

Problems:

Urban Landscaping Bids (C1–C2)

A landscaping company is bidding on city maintenance projects. The transaction is completing landscaping projects, where profit is earned per project plus a fixed profit adjustment from grant conditions.

Decision alternatives:

1. **Residential team focus** (profit 0.6 thousand USD per project)
2. **Commercial team focus** (profit 1.0 thousand USD per project)

Grant states with fixed profit components:

1. Grant available $\rightarrow +10$ thousand USD
2. No grant $\rightarrow -5$ thousand USD

Modeled projects completed:

1. Residential $\rightarrow$ 40 (Grant available), 25 (No grant)
2. Commercial $\rightarrow$ 30 (Grant available), 20 (No grant)

Questions/Tasks.

C1 Interpreting decision alternatives, events, consequences, and states.

C2 Building the payoff table.

C1 - Interpreting decision alternatives, events, consequences, and states.

Element	Description	
Alternatives	**A1:** Residential	**A2:** Commercial
Statesofnature	**S1:** Grant available	**S2:** No grant
Events	Grant outcome affecting the fixed profit adjustment	
Consequences	Profit in thousand USD = projects × profit per project + fixed profit	

C2 - Building the payoff table. Calculation table in thousand USD:

$$\text{Payoff} = \text{projects completed} \times \text{profit per project} + \text{grant fixed profit}$$

Alternative	Grant available	No grant
Residential focus	$40 \times 0.6 + 10 = 34$	$25 \times 0.6 - 5 = 10$
Commercial focus	$30 \times 1.0 + 10 = 40$	$20 \times 1.0 - 5 = 15$

Alternative	Grant available	No grant
Residential focus	34	10
Commercial focus	40	15

Solar Installation Bids (C1–C2)

A solar company is bidding on installation contracts. The transaction is installing solar systems, where profit is earned per installation plus a fixed profit adjustment from policy conditions.

Decision alternatives:

1. **Residential focus** (profit 0.9 thousand USD per installation)
2. **Commercial focus** (profit 1.6 thousand USD per installation)
3. **Mixed portfolio** (profit 1.2 thousand USD per installation)

Modeled installations:

1. Residential → 80 (Rebate continues), 50 (Rebate ends)
2. Commercial → 55 (Rebate continues), 40 (Rebate ends)
3. Mixed → 70 (Rebate continues), 45 (Rebate ends)

Policy states with fixed profit components:

1. Rebate continues → +20 thousand USD
2. Rebate ends → −10 thousand USD

Questions/Tasks.

C1 Interpreting decision alternatives, events, consequences, and states.

C2 Building the payoff table.

C1 - Interpreting decision alternatives, events, consequences, and states.

Element	Description		
Alternatives	**A1:** Residential	**A2:** Commercial	**A3:** Mixed portfolio
Statesofnature	**S1:** Rebate continues	**S2:** Rebate ends	
Events	Policy outcome affecting the fixed profit adjustment		
Consequences	Profit in thousand USD = installs × profit per install + fixed profit		

C2 - Building the payoff table. Calculation table in thousand USD:

$$\text{Payoff} = \text{installations} \times \text{profit per installation} + \text{policy fixed profit}$$

Alternative	Rebate continues	Rebate ends
Residential focus	$80 \times 0.9 + 20 = 92$	$50 \times 0.9 - 10 = 35$
Commercial focus	$55 \times 1.6 + 20 = 108$	$40 \times 1.6 - 10 = 54$
Mixed portfolio	$70 \times 1.2 + 20 = 104$	$45 \times 1.2 - 10 = 44$

Alternative	Rebate continues	Rebate ends
Residential focus	92	35
Commercial focus	108	54
Mixed portfolio	104	44

4.4 State-Based Variable Plus Alternative-Based Fixed

Regional Bus Charters (C1–C2)

A regional transport firm is choosing a charter fleet for school trips. The transaction is bus charters, where profit is earned per charter plus a fixed profit component tied to the selected fleet arrangement.

Alternatives:

1. **Use mini-bus fleet** (fixed profit 15 thousand USD)
2. **Use coach fleet** (fixed profit 25 thousand USD)

Profit per charter depends on fuel costs:

1. Regular fuel → 4 thousand USD
2. High fuel → 2 thousand USD

Fuel-cost states:

1. Regular fuel cost
2. High fuel cost

Modeled charters:

1. Mini-bus → 25 (Regular), 20 (High)
2. Coach → 18 (Regular), 16 (High)

Questions/Tasks.

C1 Interpreting decision alternatives, events, consequences, and states.

C2 Building the payoff table.

C1 - Interpreting decision alternatives, events, consequences, and states.

Element	Description	
Alternatives	**A1:** Mini-bus fleet	**A2:** Coach fleet
Statesofnature	**S1:** Regular fuel	**S2:** High fuel
Events	Fuel costs affecting profit per charter	
Consequences	Profit in thousand USD = charters × profit per charter + fixed profit	

C2 - Building the payoff table. Calculation table in thousand USD:

$$\text{Payoff} = \text{charters} \times \text{profit per charter} + \text{fleet fixed profit}$$

Alternative	Regular fuel	High fuel
Mini-bus fleet	$25 \times 4 + 15 = 115$	$20 \times 2 + 15 = 55$
Coach fleet	$18 \times 4 + 25 = 97$	$16 \times 2 + 25 = 57$

Alternative	Regular fuel	High fuel
Mini-bus fleet	115	55
Coach fleet	97	57

Air Cargo Charters (C1–C2)

An air cargo firm is choosing a charter fleet. The transaction is cargo charters, where profit is earned per charter plus a fixed profit component tied to the chosen aircraft lease.

Alternatives:

1. **Lease narrow-body aircraft** (fixed profit 30 thousand USD)
2. **Lease wide-body aircraft** (fixed profit 50 thousand USD)

Fuel-price states:

1. Low fuel cost
2. Medium fuel cost
3. High fuel cost

Profit per charter depends on fuel prices:

1. Low fuel → 8 thousand USD
2. Medium fuel → 5 thousand USD
3. High fuel → 2 thousand USD

Modeled charters:

1. Narrow-body → 20 (Low), 18 (Medium), 15 (High)
2. Wide-body → 14 (Low), 12 (Medium), 9 (High)

Questions/Tasks.

C1 Interpreting decision alternatives, events, consequences, and states.

C2 Building the payoff table.

C1 - Interpreting decision alternatives, events, consequences, and states.

Element	Description		
Alternatives	**A1:** Narrow-body lease	**A2:** Wide-body lease	
Statesofnature	**S1:** Low fuel	**S2:** Medium fuel	**S3:** High fuel
Events	Fuel prices affecting profit per charter		
Consequences	Profit in thousand USD = charters × profit per charter + fixed profit		

C2 - Building the payoff table. Calculation table in thousand USD:

$$\text{Payoff} = \text{charters} \times \text{profit per charter} + \text{aircraft fixed profit}$$

Alternative	Low fuel	Medium fuel	High fuel
Narrow-body lease	$20 \times 8 + 30 = 190$	$18 \times 5 + 30 = 120$	$15 \times 2 + 30 = 60$
Wide-body lease	$14 \times 8 + 50 = 162$	$12 \times 5 + 50 = 110$	$9 \times 2 + 50 = 68$

Alternative	Low fuel	Medium fuel	High fuel
Narrow-body lease	190	120	60
Wide-body lease	162	110	68

4.5 State-Based Variable Plus State-Based Fixed

Problems:
4.5. Fruit Export Permits (C1–C2)
4.5. Seafood Export Licenses (C1–C2)

Fruit Export Permits (C1–C2)

A fruit exporter is choosing an export permit focus. The transaction is exporting fruit crates, where profit is earned per crate plus a fixed profit adjustment tied to seasonal conditions.

Alternatives:

1. **Citrus export focus**
2. **Berry export focus**

Seasonal states with fixed profit components:

1. Strong harvest season → +12 thousand USD
2. Weak harvest season → −8 thousand USD

Profit per crate depends on the season:

1. Strong season → 0.4 thousand USD
2. Weak season → 0.2 thousand USD

Modeled crates exported:

1. Citrus → 200 (Strong), 150 (Weak)
2. Berry → 180 (Strong), 120 (Weak)

Questions/Tasks.

C1 Interpreting decision alternatives, events, consequences, and states.

C2 Building the payoff table.

C1 - Interpreting decision alternatives, events, consequences, and states.

Element	Description	
Alternatives	**A1:** Citrus focus	**A2:** Berry focus
Statesofnature	**S1:** Strong	**S2:** Weak
Events	Seasonal conditions affecting prices and fixed profit adjustments	
Consequences	Profit in thousand USD = crates × profit per crate + fixed profit	

C2 - Building the payoff table. Calculation table in thousand USD:

$$\text{Payoff} = \text{crates exported} \times \text{profit per crate} + \text{season fixed profit}$$

Alternative	Strong	Weak
Citrus focus	$200 \times 0.4 + 12 = 92$	$150 \times 0.2 - 8 = 22$
Berry focus	$180 \times 0.4 + 12 = 84$	$120 \times 0.2 - 8 = 16$

Alternative	Strong	Weak
Citrus focus	92	22
Berry focus	84	16

Seafood Export Licenses (C1–C2)

A seafood exporter is choosing a license focus. The transaction is exporting seafood crates, where profit is earned per crate plus a fixed profit adjustment tied to seasonal conditions.

Alternatives:

1. **Frozen fillet focus**
2. **Fresh export focus**

Seasonal states with fixed profit components:

1. Strong tourism season $\rightarrow +30$ thousand USD
2. Normal season $\rightarrow +10$ thousand USD
3. Stormy season $\rightarrow -15$ thousand USD

Profit per crate depends on the season:

1. Strong $\rightarrow 0.7$ thousand USD
2. Normal $\rightarrow 0.5$ thousand USD
3. Stormy $\rightarrow 0.2$ thousand USD

Modeled crates exported:

1. Frozen fillet $\rightarrow$ 120 (Strong), 100 (Normal), 80 (Stormy)
2. Fresh export $\rightarrow$ 90 (Strong), 110 (Normal), 70 (Stormy)

Questions/Tasks.

C1 Interpreting decision alternatives, events, consequences, and states.

C2 Building the payoff table.

C1 - Interpreting decision alternatives, events, consequences, and states.

Element	Description		
Alternatives	**A1:** Frozen fillet focus	**A2:** Fresh export focus	
Statesofnature	**S1:** Strong	**S2:** Normal	**S3:** Stormy
Events	Seasonal conditions affecting prices and fixed profit adjustments		
Consequences	Profit in thousand USD = crates $\times$ profit per crate + fixed profit		

C2 - Building the payoff table. Calculation table in thousand USD:

$$\text{Payoff} = \text{crates exported} \times \text{profit per crate} + \text{season fixed profit}$$

Alternative	Strong	Normal	Stormy
Frozen fillet focus	$120 \times 0.7 + 30 = 114$	$100 \times 0.5 + 10 = 60$	$80 \times 0.2 - 15 = 1$
Fresh export focus	$90 \times 0.7 + 30 = 93$	$110 \times 0.5 + 10 = 65$	$70 \times 0.2 - 15 = -1$

Alternative	Strong	Normal	Stormy
Frozen fillet focus	114	60	1
Fresh export focus	93	65	-1

4.6 Alternative-Based Fixed Plus State-Based Fixed

Problems:

4.6. Warehouse Security Agreements (C1–C2)

4.6. Harbor Inspection Agreements (C1–C2)

Warehouse Security Agreements (C1–C2)

A warehouse operator is choosing a security agreement for seasonal inventory protection. The transaction is a fixed security agreement, where the recorded payoff combines a fixed component tied to the agreement and a fixed component tied to the demand season.

Decision alternatives with fixed payoff components:

1. **Standard security agreement** $\rightarrow$ 20 thousand USD
2. **Enhanced security agreement** $\rightarrow$ 35 thousand USD

Demand-season states with fixed payoff components:

1. Normal season $\rightarrow$ 6 thousand USD
2. Festival season $\rightarrow$ 14 thousand USD

Questions/Tasks.

C1 Interpreting decision alternatives, events, consequences, and states.

C2 Building the payoff table.

C1 - Interpreting decision alternatives, events, consequences, and states.

Element	Description	
Alternatives	**A1:** Standard security	**A2:** Enhanced security
Statesofnature	**S1:** Normal season	**S2:** Festival season
Events	Demand-season state realized during the security period	
Consequences	Profit in thousand USD = agreement fixed payoff + state fixed payoff	

C2 - Building the payoff table. Calculation table in thousand USD:

$$\text{Payoff} = \text{agreement fixed payoff} + \text{state fixed payoff}$$

Alternative	Normal season	Festival season
Standard security	$20 + 6 = 26$	$20 + 14 = 34$
Enhanced security	$35 + 6 = 41$	$35 + 14 = 49$

Alternative	Normal season	Festival season
Standard security	26	34
Enhanced security	41	49

Harbor Inspection Agreements (C1–C2)

A port authority is choosing an inspection agreement for harbor facilities. The transaction is a fixed inspection agreement, where the recorded payoff combines a fixed component tied to the agreement and a fixed component tied to the harbor condition state.

Decision alternatives with fixed payoff components:

1. **Standard inspection agreement** → 28 thousand USD
2. **Rapid-response inspection agreement** → 42 thousand USD

Harbor condition states with fixed payoff components:

1. Calm season → 12 thousand USD
2. Busy season → 5 thousand USD
3. Storm-repair season → −10 thousand USD

Questions/Tasks.

C1 Interpreting decision alternatives, events, consequences, and states.

C2 Building the payoff table.

C1 - Interpreting decision alternatives, events, consequences, and states.

Element	Description		
Alternatives	**A1:** Standard inspection	**A2:** Rapid-response inspection	
Statesofnature	**S1:** Calm season	**S2:** Busy season	**S3:** Storm-repair season
Events	Harbor condition state realized during the inspection period		
Consequences	Profit in thousand USD = agreement fixed payoff + state fixed payoff		

C2 - Building the payoff table. Calculation table in thousand USD:

$$\text{Payoff} = \text{agreement fixed payoff} + \text{state fixed payoff}$$

Alternative	Calm season	Busy season	Storm-repair season
Standard inspection	$28 + 12 = 40$	$28 + 5 = 33$	$28 - 10 = 18$
Rapid-response inspection	$42 + 12 = 54$	$42 + 5 = 47$	$42 - 10 = 32$

Alternative	Calm season	Busy season	Storm-repair season
Standard inspection	40	33	18
Rapid-response inspection	54	47	32

5 Three-Component Models

Problems:

5.1 Alternative-Based Variable Plus State-Based Variable Plus Alternative-Based Fixed

Problems:

Water Treatment Modules (C1–C2)

A utility chooses a portable treatment design for batches of water delivered to remote communities.

Decision alternatives

1. Membrane unit
2. Filtration unit

States of nature

1. Dry season
2. Wet season

Alternative-based variable payoff per item $v(A_i)$

1. Membrane unit: 0.8 thousand USD
2. Filtration unit: 0.6 thousand USD

State-based variable payoff per item $v(S_j)$

1. Dry season: 0.3 thousand USD
2. Wet season: 0.1 thousand USD

Alternative-based fixed payoff $f(A_i)$

1. Membrane unit: 8 thousand USD
2. Filtration unit: 12 thousand USD

Modeled number of items $q(A_i, S_j)$

1. Membrane unit: 50 under Dry season, 35 under Wet season
2. Filtration unit: 60 under Dry season, 45 under Wet season

Questions/Tasks.

C1 Interpreting decision alternatives, events, consequences, and states.

C2 Building the payoff table.

C1 - Interpreting decision alternatives, events, consequences, and states.

Element	Description	
Alternatives	**A1:** Membrane unit	**A2:** Filtration unit
Statesofnature	**S1:** Dry season	**S2:** Wet season
Events	The realized state affecting the transaction	
Consequences	Payoff in thousand USD for each alternative–state pair	

C2 - Building the payoff table. The applicable payoff formula is

$$P_{ay}(A_i, S_j) = q(A_i, S_j)\,(v(A_i) + v(S_j)) + f(A_i).$$

Detailed calculations (thousand USD):

Alternative	Dry season	Wet season
Membrane unit	$50 \times (0.8 + 0.3) + 8 = 63$	$35 \times (0.8 + 0.1) + 8 = 39.5$
Filtration unit	$60 \times (0.6 + 0.3) + 12 = 66$	$45 \times (0.6 + 0.1) + 12 = 43.5$

The simplified payoff table is

Alternative	Dry season	Wet season
Membrane unit	63	39.5
Filtration unit	66	43.5

Freight Locker Network (C1–C2)

A parcel operator chooses a locker network for shipment batches handled under two demand states.

Decision alternatives

1. Metro lockers
2. Suburban lockers
3. Mixed network

States of nature

1. Strong demand
2. Steady demand

Alternative-based variable payoff per item $v(A_i)$

1. Metro lockers: 0.5 thousand USD
2. Suburban lockers: 0.4 thousand USD
3. Mixed network: 0.5 thousand USD

State-based variable payoff per item $v(S_j)$

1. Strong demand: 0.2 thousand USD
2. Steady demand: 0.1 thousand USD

Alternative-based fixed payoff $f(A_i)$

1. Metro lockers: 10 thousand USD
2. Suburban lockers: 14 thousand USD
3. Mixed network: 12 thousand USD

Modeled number of items $q(A_i, S_j)$

1. Metro lockers: 80 under Strong demand, 55 under Steady demand
2. Suburban lockers: 65 under Strong demand, 60 under Steady demand
3. Mixed network: 75 under Strong demand, 65 under Steady demand

Questions/Tasks.

C1 Interpreting decision alternatives, events, consequences, and states.

C2 Building the payoff table.

C1 - Interpreting decision alternatives, events, consequences, and states.

Element	Description		
Alternatives	**A1:** Metro lockers	**A2:** Suburban lockers	**A3:** Mixed network
Statesofnature	**S1:** Strong demand	**S2:** Steady demand	
Events	The realized state affecting the transaction		
Consequences	Payoff in thousand USD for each alternative–state pair		

C2 - Building the payoff table. The applicable payoff formula is

$$P_{ay}(A_i, S_j) = q(A_i, S_j)\,(v(A_i) + v(S_j)) + f(A_i).$$

Detailed calculations (thousand USD):

Alternative	Strong demand	Steady demand
Metro lockers	$80 \times (0.5 + 0.2) + 10 = 66$	$55 \times (0.5 + 0.1) + 10 = 43$
Suburban lockers	$65 \times (0.4 + 0.2) + 14 = 53$	$60 \times (0.4 + 0.1) + 14 = 44$
Mixed network	$75 \times (0.5 + 0.2) + 12 = 64.5$	$65 \times (0.5 + 0.1) + 12 = 51$

The simplified payoff table is

Alternative	Strong demand	Steady demand
Metro lockers	66	43
Suburban lockers	53	44
Mixed network	64.5	51

5.2 Alternative-Based Variable Plus State-Based Variable Plus State-Based Fixed

Problems:

District Heating Supply (C1–C2)

A district energy provider selects a heat source for units supplied during the winter.

Decision alternatives

1. Biomass heat
2. Heat pumps

States of nature

1. Mild winter
2. Cold winter

Alternative-based variable payoff per item $v(A_i)$

1. Biomass heat: 0.5 thousand USD
2. Heat pumps: 0.6 thousand USD

State-based variable payoff per item $v(S_j)$

1. Mild winter: 0.1 thousand USD
2. Cold winter: 0.3 thousand USD

State-based fixed payoff $f(S_j)$

1. Mild winter: 6 thousand USD

2. Cold winter: -4 thousand USD (negative adjustment)

Modeled number of items $q(A_i, S_j)$

1. Biomass heat: 55 under Mild winter, 85 under Cold winter
2. Heat pumps: 65 under Mild winter, 95 under Cold winter

Questions/Tasks.

C1 Interpreting decision alternatives, events, consequences, and states.

C2 Building the payoff table.

C1 - Interpreting decision alternatives, events, consequences, and states.

Element	Description	
Alternatives	**A1:** Biomass heat	**A2:** Heat pumps
Statesofnature	**S1:** Mild winter	**S2:** Cold winter
Events	The realized state affecting the transaction	
Consequences	Payoff in thousand USD for each alternative–state pair	

C2 - Building the payoff table. The applicable payoff formula is

$$P_{ay}(A_i, S_j) = q(A_i, S_j)\,(v(A_i) + v(S_j)) + f(S_j).$$

Detailed calculations (thousand USD):

Alternative	Mild winter	Cold winter
Biomass heat	$55 \times (0.5 + 0.1) + 6 = 39$	$85 \times (0.5 + 0.3) - 4 = 64$
Heat pumps	$65 \times (0.6 + 0.1) + 6 = 51.5$	$95 \times (0.6 + 0.3) - 4 = 81.5$

The simplified payoff table is

Alternative	Mild winter	Cold winter
Biomass heat	39	64
Heat pumps	51.5	81.5

Port Cold-Chain Service (C1–C2)

A port authority chooses a cold-chain service model for container batches under three operating states.

Decision alternatives

1. Automated service
2. Crew service

States of nature

1. Clear flow
2. Congestion
3. Inspection surge

Alternative-based variable payoff per item $v(A_i)$

1. Automated service: 0.7 thousand USD
2. Crew service: 0.5 thousand USD

State-based variable payoff per item $v(S_j)$

1. Clear flow: 0.2 thousand USD
2. Congestion: 0.1 thousand USD
3. Inspection surge: 0.3 thousand USD

State-based fixed payoff $f(S_j)$

1. Clear flow: 8 thousand USD
2. Congestion: -3 thousand USD (negative adjustment)
3. Inspection surge: 5 thousand USD

Modeled number of items $q(A_i, S_j)$

1. Automated service: 80 under Clear flow, 60 under Congestion, 45 under Inspection surge
2. Crew service: 70 under Clear flow, 65 under Congestion, 55 under Inspection surge

Questions/Tasks.

C1 Interpreting decision alternatives, events, consequences, and states.

C2 Building the payoff table.

C1 - Interpreting decision alternatives, events, consequences, and states.

Element	Description		
Alternatives	**A1:** Automated service	**A2:** Crew service	
Statesofnature	**S1:** Clear flow	**S2:** Congestion	**S3:** Inspection surge
Events	The realized state affecting the transaction		
Consequences	Payoff in thousand USD for each alternative–state pair		

C2 - Building the payoff table. The applicable payoff formula is

$$P_{ay}(A_i, S_j) = q(A_i, S_j)\,(v(A_i) + v(S_j)) + f(S_j).$$

Detailed calculations (thousand USD):

Alternative	Clear flow	Congestion	Inspection surge
Automated service	$80 \times (0.7 + 0.2) + 8 = 80$	$60 \times (0.7 + 0.1) - 3 = 45$	$45 \times (0.7 + 0.3) + 5 = 50$
Crew service	$70 \times (0.5 + 0.2) + 8 = 57$	$65 \times (0.5 + 0.1) - 3 = 36$	$55 \times (0.5 + 0.3) + 5 = 49$

The simplified payoff table is

Alternative	Clear flow	Congestion	Inspection surge
Automated service	80	45	50
Crew service	57	36	49

5.3 Alternative-Based Variable Plus Alternative-Based Fixed Plus State-Based Fixed

Problems:

Pharmacy Delivery Plan (C1–C2)

A pharmacy cooperative chooses a delivery plan for prescription batches served across two traffic states.

Decision alternatives

1. Cargo bikes
2. Electric vans

States of nature

1. Light traffic
2. Heavy traffic

Alternative-based variable payoff per item $v(A_i)$

1. Cargo bikes: 0.6 thousand USD
2. Electric vans: 0.8 thousand USD

Alternative-based fixed payoff $f(A_i)$

1. Cargo bikes: 9 thousand USD
2. Electric vans: 4 thousand USD

State-based fixed payoff $f(S_j)$

1. Light traffic: 6 thousand USD
2. Heavy traffic: -5 thousand USD (negative adjustment)

Modeled number of items $q(A_i, S_j)$

1. Cargo bikes: 60 under Light traffic, 45 under Heavy traffic
2. Electric vans: 75 under Light traffic, 55 under Heavy traffic

Questions/Tasks.

C1 Interpreting decision alternatives, events, consequences, and states.

C2 Building the payoff table.

C1 - Interpreting decision alternatives, events, consequences, and states.

Element	Description	
Alternatives	**A1:** Cargo bikes	**A2:** Electric vans
Statesofnature	**S1:** Light traffic	**S2:** Heavy traffic
Events	The realized state affecting the transaction	
Consequences	Payoff in thousand USD for each alternative–state pair	

C2 - Building the payoff table. The applicable payoff formula is

$$P_{ay}(A_i, S_j) = q(A_i, S_j)v(A_i) + f(A_i) + f(S_j).$$

Detailed calculations (thousand USD):

Alternative	Light traffic	Heavy traffic
Cargo bikes	$60 \times 0.6 + 9 + 6 = 51$	$45 \times 0.6 + 9 - 5 = 31$
Electric vans	$75 \times 0.8 + 4 + 6 = 70$	$55 \times 0.8 + 4 - 5 = 43$

The simplified payoff table is

Alternative	Light traffic	Heavy traffic
Cargo bikes	51	31
Electric vans	70	43

Timber Processing Line (C1–C2)

A wood cooperative selects a processing line for lumber batches sold under three market conditions.

Decision alternatives

1. Precision line
2. Standard line

States of nature

1. Firm market
2. Normal market
3. Weak market

Alternative-based variable payoff per item $v(A_i)$

1. Precision line: 0.9 thousand USD
2. Standard line: 0.6 thousand USD

Alternative-based fixed payoff $f(A_i)$

1. Precision line: 7 thousand USD
2. Standard line: 13 thousand USD

State-based fixed payoff $f(S_j)$

1. Firm market: 10 thousand USD
2. Normal market: 2 thousand USD
3. Weak market: -6 thousand USD (negative adjustment)

Modeled number of items $q(A_i, S_j)$

1. Precision line: 70 under Firm market, 55 under Normal market, 40 under Weak market
2. Standard line: 85 under Firm market, 65 under Normal market, 50 under Weak market

Questions/Tasks.

C1 Interpreting decision alternatives, events, consequences, and states.

C2 Building the payoff table.

C1 - Interpreting decision alternatives, events, consequences, and states.

Element	Description		
Alternatives	**A1:** Precision line	**A2:** Standard line	
Statesofnature	**S1:** Firm market	**S2:** Normal market	**S3:** Weak market
Events	The realized state affecting the transaction		
Consequences	Payoff in thousand USD for each alternative–state pair		

C2 - Building the payoff table. The applicable payoff formula is

$$P_{ay}(A_i, S_j) = q(A_i, S_j)v(A_i) + f(A_i) + f(S_j).$$

Detailed calculations (thousand USD):

Alternative	Firm market	Normal market	Weak market
Precision line	$70 \times 0.9 + 7 + 10 = 80$	$55 \times 0.9 + 7 + 2 = 58.5$	$40 \times 0.9 + 7 - 6 = 37$
Standard line	$85 \times 0.6 + 13 + 10 = 74$	$65 \times 0.6 + 13 + 2 = 54$	$50 \times 0.6 + 13 - 6 = 37$

The simplified payoff table is

Alternative	Firm market	Normal market	Weak market
Precision line	80	58.5	37
Standard line	74	54	37

5.4 State-Based Variable Plus Alternative-Based Fixed Plus State-Based Fixed

Problems:

Public Library Outreach (C1–C2)

A library system selects an outreach format for participant groups served under two attendance states.

Decision alternatives

1. Mobile library
2. Pop-up branches

States of nature

1. High attendance
2. Low attendance

State-based variable payoff per item $v(S_j)$

1. High attendance: 0.7 thousand USD
2. Low attendance: 0.4 thousand USD

Alternative-based fixed payoff $f(A_i)$

1. Mobile library: 8 thousand USD
2. Pop-up branches: 11 thousand USD

State-based fixed payoff $f(S_j)$

1. High attendance: 5 thousand USD
2. Low attendance: -2 thousand USD (negative adjustment)

Modeled number of items $q(A_i, S_j)$

1. Mobile library: 50 under High attendance, 30 under Low attendance

2. Pop-up branches: 45 under High attendance, 35 under Low attendance

Questions/Tasks.

C1 Interpreting decision alternatives, events, consequences, and states.

C2 Building the payoff table.

C1 - Interpreting decision alternatives, events, consequences, and states.

Element	Description	
Alternatives	**A1:** Mobile library	**A2:** Pop-up branches
Statesofnature	**S1:** High attendance	**S2:** Low attendance
Events	The realized state affecting the transaction	
Consequences	Payoff in thousand USD for each alternative–state pair	

C2 - Building the payoff table. The applicable payoff formula is

$$P_{ay}(A_i, S_j) = q(A_i, S_j)v(S_j) + f(A_i) + f(S_j).$$

Detailed calculations (thousand USD):

Alternative	High attendance	Low attendance
Mobile library	$50 \times 0.7 + 8 + 5 = 48$	$30 \times 0.4 + 8 - 2 = 18$
Pop-up branches	$45 \times 0.7 + 11 + 5 = 47.5$	$35 \times 0.4 + 11 - 2 = 23$

The simplified payoff table is

Alternative	High attendance	Low attendance
Mobile library	48	18
Pop-up branches	47.5	23

Grain Terminal Schedule (C1–C2)

A grain terminal selects a loading schedule for cargo lots handled under two export states.

Decision alternatives

1. Day schedule
2. Night schedule
3. Split schedule

States of nature

1. Open channels
2. Restricted channels

State-based variable payoff per item $v(S_j)$

1. Open channels: 0.6 thousand USD
2. Restricted channels: 0.3 thousand USD

Alternative-based fixed payoff $f(A_i)$

1. Day schedule: 12 thousand USD
2. Night schedule: 8 thousand USD
3. Split schedule: 10 thousand USD

State-based fixed payoff $f(S_j)$

1. Open channels: 7 thousand USD
2. Restricted channels: -3 thousand USD (negative adjustment)

Modeled number of items $q(A_i, S_j)$

1. Day schedule: 90 under Open channels, 55 under Restricted channels
2. Night schedule: 75 under Open channels, 60 under Restricted channels
3. Split schedule: 85 under Open channels, 65 under Restricted channels

Questions/Tasks.

C1 Interpreting decision alternatives, events, consequences, and states.

C2 Building the payoff table.

C1 - Interpreting decision alternatives, events, consequences, and states.

Element	Description		
Alternatives	**A1:** Day schedule	**A2:** Night schedule	**A3:** Split schedule
Statesofnature	**S1:** Open channels	**S2:** Restricted channels	
Events	The realized state affecting the transaction		
Consequences	Payoff in thousand USD for each alternative–state pair		

C2 - Building the payoff table. The applicable payoff formula is

$$P_{ay}(A_i, S_j) = q(A_i, S_j)v(S_j) + f(A_i) + f(S_j).$$

Detailed calculations (thousand USD):

Alternative	Open channels	Restricted channels
Day schedule	$90 \times 0.6 + 12 + 7 = 73$	$55 \times 0.3 + 12 - 3 = 25.5$
Night schedule	$75 \times 0.6 + 8 + 7 = 60$	$60 \times 0.3 + 8 - 3 = 23$
Split schedule	$85 \times 0.6 + 10 + 7 = 68$	$65 \times 0.3 + 10 - 3 = 26.5$

The simplified payoff table is

Alternative	Open channels	Restricted channels
Day schedule	73	25.5
Night schedule	60	23
Split schedule	68	26.5

6 Complete Four-Component Models

Problems:

6.1 Alternative-Based Variable Plus State-Based Variable Plus Alternative-Based Fixed Plus State-Based Fixed

This structure keeps all four payoff components explicit: an alternative-based variable payoff per item $v(A_i)$, a state-based variable payoff per item $v(S_j)$, an alternative-based fixed payoff $f(A_i)$, and a state-based fixed payoff $f(S_j)$. The quantity $q(A_i, S_j)$ may vary for every alternative–state pair, so

$$P_{ay}(A_i, S_j) = q(A_i, S_j)\,(v(A_i) + v(S_j)) + f(A_i) + f(S_j).$$

Problems:

6.1. Urban Food Hub (C1–C2)
6.1. Regional Rail Cargo (C1–C2)

Urban Food Hub (C1–C2)

A city food hub chooses a distribution system for produce crates delivered under two demand states.

Decision alternatives

1. Electric sorting
2. Manual sorting

States of nature

1. Strong demand
2. Soft demand

Alternative-based variable payoff per item $v(A_i)$

1. Electric sorting: 0.6 thousand USD
2. Manual sorting: 0.4 thousand USD

State-based variable payoff per item $v(S_j)$

1. Strong demand: 0.2 thousand USD
2. Soft demand: 0.1 thousand USD

Alternative-based fixed payoff $f(A_i)$

1. Electric sorting: 9 thousand USD
2. Manual sorting: 14 thousand USD

State-based fixed payoff $f(S_j)$

1. Strong demand: 6 thousand USD
2. Soft demand: -3 thousand USD (negative adjustment)

Modeled number of items $q(A_i, S_j)$

1. Electric sorting: 70 under Strong demand, 45 under Soft demand
2. Manual sorting: 80 under Strong demand, 55 under Soft demand

Questions/Tasks.

C1 Interpreting decision alternatives, events, consequences, and states.

C2 Building the payoff table.

C1 - Interpreting decision alternatives, events, consequences, and states.

Element	Description	
Alternatives	**A1:** Electric sorting	**A2:** Manual sorting
Statesofnature	**S1:** Strong demand	**S2:** Soft demand
Events	The realized state affecting the transaction	
Consequences	Payoff in thousand USD for each alternative–state pair	

C2 - Building the payoff table. The applicable payoff formula is

$$P_{ay}(A_i, S_j) = q(A_i, S_j)\,(v(A_i) + v(S_j)) + f(A_i) + f(S_j).$$

Detailed calculations (thousand USD):

Alternative	Strong demand	Soft demand
Electric sorting	$70 \times (0.6 + 0.2) + 9 + 6 = 71$	$45 \times (0.6 + 0.1) + 9 - 3 = 37.5$
Manual sorting	$80 \times (0.4 + 0.2) + 14 + 6 = 68$	$55 \times (0.4 + 0.1) + 14 - 3 = 38.5$

The simplified payoff table is

Alternative	Strong demand	Soft demand
Electric sorting	71	37.5
Manual sorting	68	38.5

Regional Rail Cargo (C1–C2)

A rail operator selects a cargo service design for freight lots moved under three network states.

Decision alternatives

1. Express service
2. Flexible service

States of nature

1. Clear network
2. Busy network
3. Disrupted network

Alternative-based variable payoff per item $v(A_i)$

1. Express service: 0.8 thousand USD
2. Flexible service: 0.6 thousand USD

State-based variable payoff per item $v(S_j)$

1. Clear network: 0.2 thousand USD
2. Busy network: 0.1 thousand USD
3. Disrupted network: -0.1 thousand USD (negative adjustment)

Alternative-based fixed payoff $f(A_i)$

1. Express service: 10 thousand USD
2. Flexible service: 15 thousand USD

State-based fixed payoff $f(S_j)$

1. Clear network: 5 thousand USD
2. Busy network: 0 thousand USD
3. Disrupted network: -7 thousand USD (negative adjustment)

1. Express service: 85 under Clear network, 65 under Busy network, 40 under Disrupted network
2. Flexible service: 75 under Clear network, 70 under Busy network, 50 under Disrupted network

Modeled number of items $q(A_i, S_j)$

Questions/Tasks.

C1 Interpreting decision alternatives, events, consequences, and states.

C2 Building the payoff table.

C1 - Interpreting decision alternatives, events, consequences, and states.

Element	Description		
Alternatives	**A1:** Express service	**A2:** Flexible service	
Statesofnature	**S1:** Clear network	**S2:** Busy network	**S3:** Disrupted network
Events	The realized state affecting the transaction		
Consequences	Payoff in thousand USD for each alternative–state pair		

C2 - Building the payoff table. The applicable payoff formula is

$$P_{ay}(A_i, S_j) = q(A_i, S_j)\,(v(A_i) + v(S_j)) + f(A_i) + f(S_j).$$

Detailed calculations (thousand USD):

Alternative	Clear network	Busy network	Disrupted network
Express service	$85 \times (0.8 + 0.2) + 10 + 5 = 100$	$65 \times (0.8 + 0.1) + 10 + 0 = 68.5$	$40 \times (0.8 - 0.1) + 10 - 7 = 31$
Flexible service	$75 \times (0.6 + 0.2) + 15 + 5 = 80$	$70 \times (0.6 + 0.1) + 15 + 0 = 64$	$50 \times (0.6 - 0.1) + 15 - 7 = 33$

The simplified payoff table is

Alternative	Clear network	Busy network	Disrupted network
Express service	100	68.5	31
Flexible service	80	64	33